Business Management and Training

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UE 8 Private Sector & Development

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Outline

Modeling exercise discussion

Motivation: Variation in firm productivity

Business management and productivity

Bloom et al (2013): Does management matter?

Adoption of management practices

Student presentations

Wrapping up

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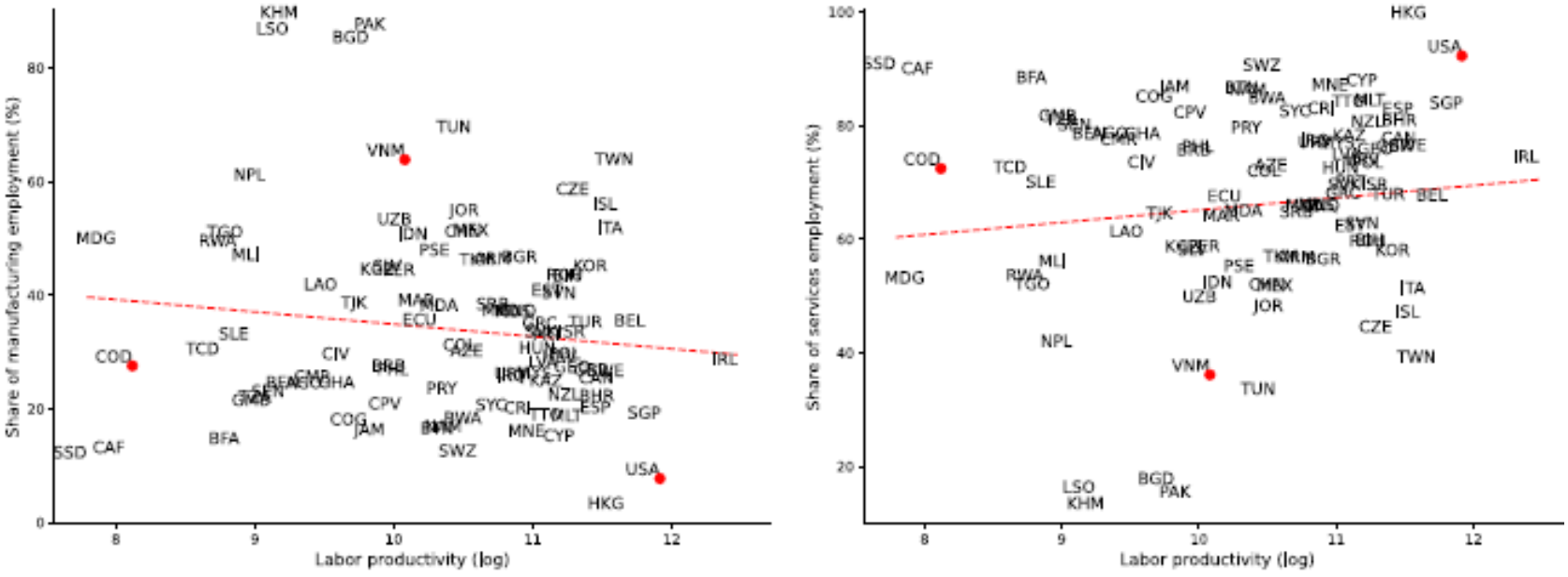
Differences in labor productivity across countries

Figure 2: Shares of Employment in Manufacturing and Services, WBES



(a) Manufacturing (b) Services

Source: Restuccia 2025

Differences in agricultural productivity across countries

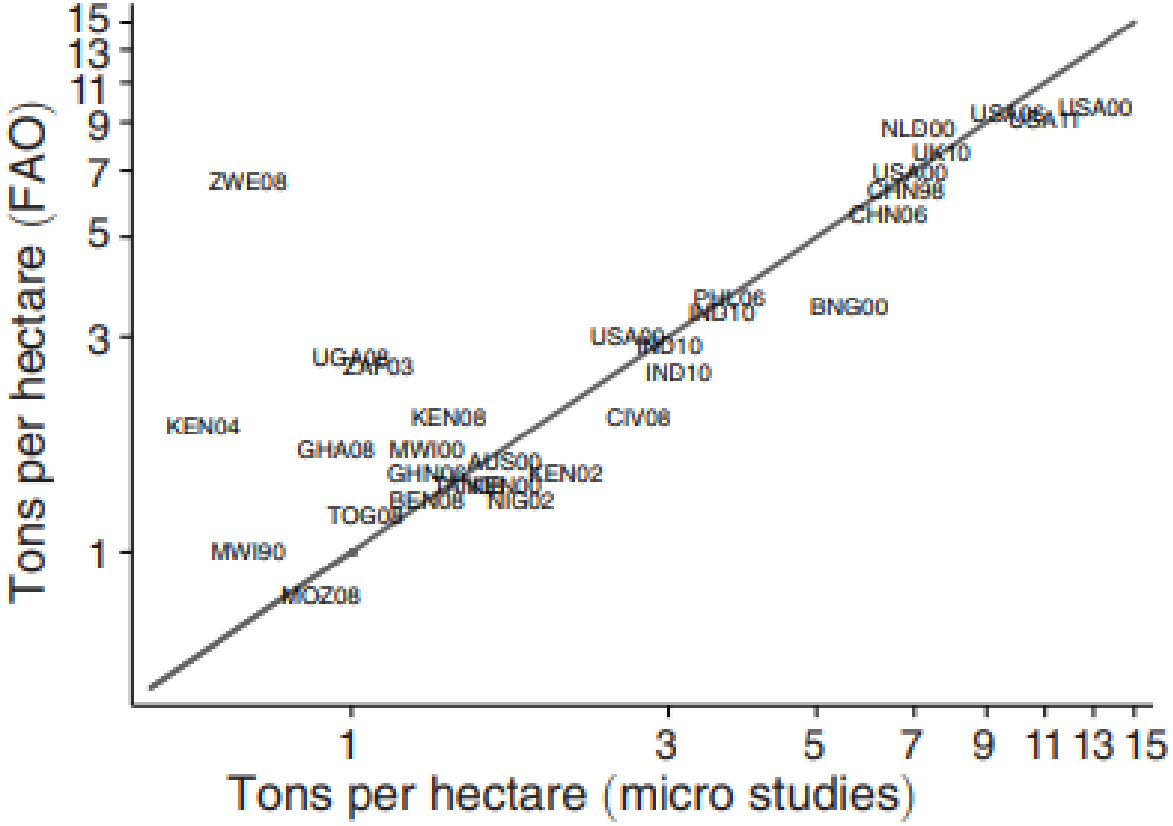


FIGURE 1. GRAIN YIELDS FROM AGGREGATE AND MICRO SOURCES

Source: Gollin, Lagakos, & Waugh 2014

Differences in sectoral productivity *within* countries

TABLE 1—SECTOR LABOR PRODUCTIVITY DIFFERENCES AND EMPLOYMENT SHARES

	Agriculture	Aggregate	Non-agriculture	Ag/non-agriculture ratio
90–10 labor productivity difference	45	22	4	10.7
Employment shares ninetieth percentile country	3		97	
Employment shares tenth percentile country	78		22	

Notes: The aggregate productivity difference is the ratio of GDP per worker between the ninetieth and tenth percentile countries. Sector productivity differences are the ratio of sector output per worker in the ninetieth and tenth percentile countries. The Ag/Non-agriculture Ratio is the agriculture productivity difference divided by the non-agriculture productivity difference.

Source: Caselli (2005).

Source: Lagakos & Waugh 2013

- In cross-section, larger dispersion of TFP is a sign of misallocation (Hsieh & Klenow 2012, Bartlesman et al 2013)
- Why?

Differences in manufacturing productivity dispersion across countries

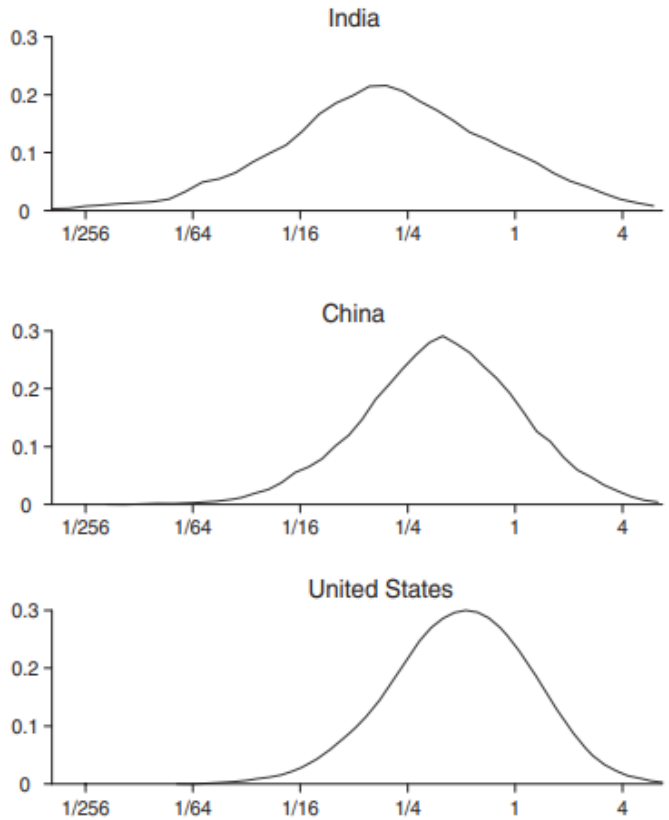


FIGURE I
Distribution of TFPQ

Source: Hsieh & Klenow 2009

TFPQ: measure of physical productivity accounting for plant-specific prices

What can explain TFP dispersion in developing countries?

Many potential factors related to topics covered in this course

- Financial market failures
- Labor market failures
- Institutional constraints: rule of law, corruption, etc.
- Social constraints
- Psychological/cognitive constraints
- Human capital constraints
  - Today: firm management and training

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How can management practices affect business productivity?

How can management practices affect business productivity?

Some examples:

- Formal record-keeping: shifting from memory-based tracking to ledger or digital systems to monitor cash flow and identify shrinkage (theft or waste)
- Inventory management: implementing basic stock-tracking to identify demand trends and prevent frequent stockouts
- Standardized operations: creating clear protocols for tasks to reduce defect rates and material waste
- Target setting and monitoring: setting daily or weekly production goals with regular feedback
- Incentive-based pay: replacing flat daily wages with piece-rate or bonus structures to align worker effort with firm output
- Employee mentoring and training: create opportunities for worker growth to promote retention

Stylized facts on management in developing countries

- Wide dispersion in management quality: much higher variance in management scores
- Long tail of poorly managed firms that persist despite inefficiency.
- Information and perception gaps: overestimation of firm's management quality
- Correlation with firm ownership: more family-managed firms, passed down generations
- Size and management quality: small firms often lack the resources to implement formal systems
- Lower levels of market competition in many developing contexts: allow poorly managed firms to survive without adopting modern practices

Constraint to productivity: Adoption of business and management practices

- Wide range of business and management practices most firms could benefit from adopting
  - Examples: separating household and business accounts, keeping basic records, monitoring inventory, budgeting, quality control, employee hiring and retention, etc.
- Particular needs for small firms in developing countries
- Evidence that firms adopting such practices are more productive and grow faster (Bloom & van Reenen 2010, McKenzie & Woodruff 2017)

Evidence on returns to business training

- $>$ \$1 billion spent annually on training at least 4-5 million potential and existing entrepreneurs in developing countries
- Growing evidence base on training interventions (summarized in McKenzie & Woodruff VoxDevLit)
- Key takeaways Gaps
  1. Entrepreneurship can be taught
  2. Better business practices matter for all sizes of enterprises
  3. Training can at least modestly improve business practices for microenterprises
  4. More intense consulting improves performance of larger firms and their subsequent growth
  5. Innovating the content and delivery methods for training is important
- Update of McKenzie (2021) meta-analysis
  - Training on average increases sales by 5.6% and profits by 12.1%
  - Some heterogeneity and power limitations across studies

Many types of business training

- Traditional entrepreneurship training More
  - Classroom style, basic business practices
  - Modest but significant average effects on improving business practices and outcomes in many settings
- Personal initiative and heuristic training More
  - Entrepreneurial mindset, rules of thumb
  - Effective for subsistence entrepreneurs if trainers are high-quality
- Consulting (Today: Bloom et al 2013) More
  - Customized, intensive, individually-tailored training
  - Evidence of effectiveness for medium/large firms
- Online training and consulting More
  - Increasingly used, scalable, low-cost, increasingly accesible
  - Evidence of modest benefits, especially if paired with some personal training
- Others Kaizen Mentoring Incubators/Accelerators

Bloom et al research agenda: role of firm management

1. To what extent can differences in management practices account for differences in TFP across firms and countries?
   - Management practices differ within and across countries and are strongly associated with firm productivity, profitability, and survival (Bloom et al 2007)
   - Management practices account for more than 20% of variation of productivity in US plants; similar share to role of R&D, ICT, human capital (Bloom et al 2019)
2. How do management practices enter the production function?
   - Better-managed firms recruit and retain workers with higher average human capital (Bender et al 2018)
   - Social capital as proxied by trust affects internal firm organization (e.g., decentralization, delegation) and thus aggregate productivity (Bloom et al 2012)
   - Causal evidence: Bloom et al (2013) $\Rightarrow$ today

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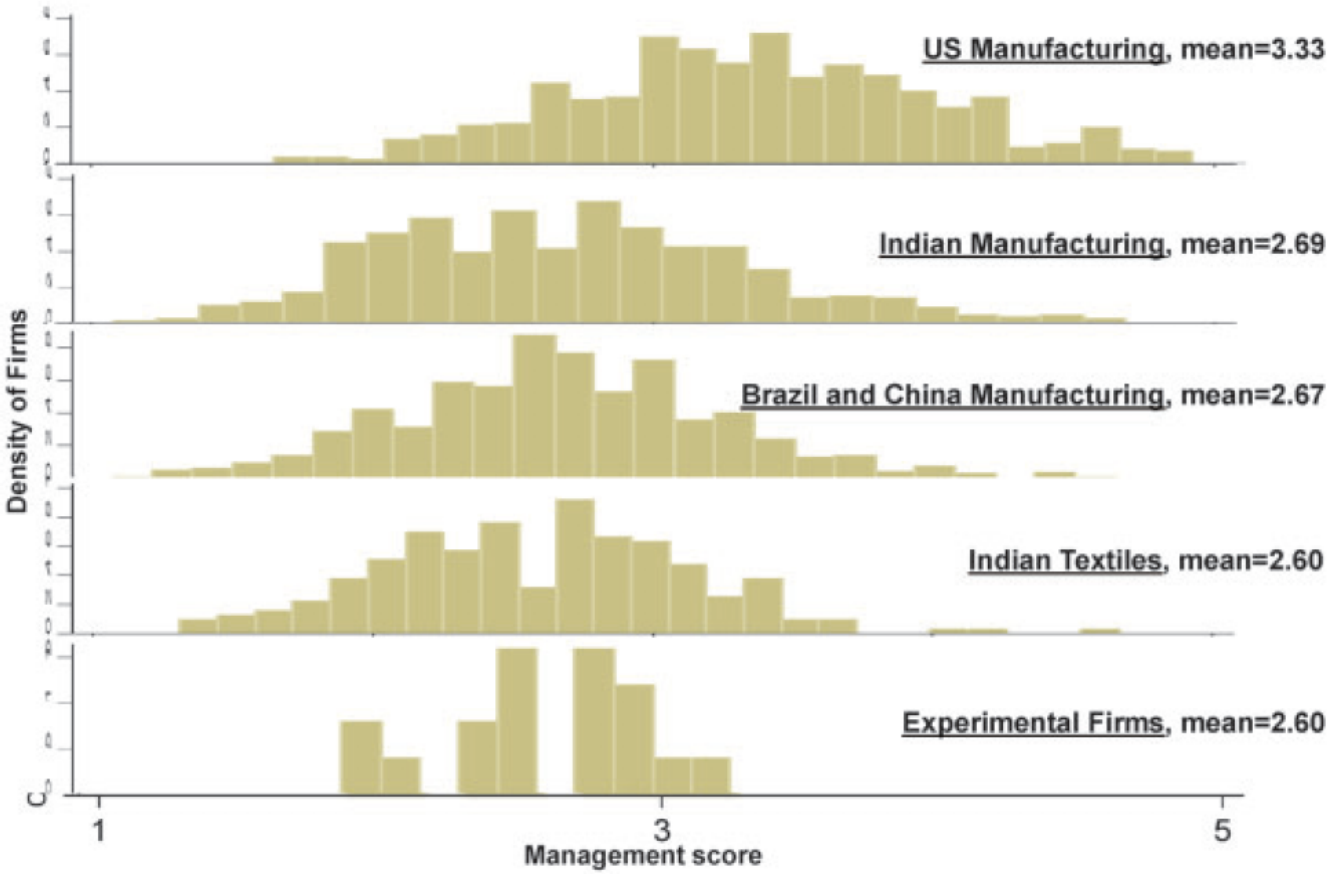
Motivation: Variation in firm management quality

- Large productivity dispersion in developing countries
- Firm management score spreads look similar to TFP spreads
- Large literature on correlations between management practices and productivity, but no causal evidence before Bloom et al ()2013)



Motivation: Observably poor management practices

Garbage outside a factory

Garbage inside a factory

Garbage inside a factory

Shelves overfilled and disorganized

Motivation: Observably poor management practices

Paper summary

- Design: First experimental evidence on the importance of management practices in large firms; sample of large multiplant Indian textile firms
  - Random assignment of extensive management consulting to adopt recommended improved management practices
- Results: Treated plants have significant improvements in quality, inventory, and output, and estimated productivity increase of 17% in the first year
  - Suggestive evidence that better management allowed greater delegation and enabled firms to open more production plants in three years after experiment
  - Adoption of improved management practices by other plants owned by firm
- Mechanisms
  - Information constraints appear to be most important factor for low initial adoption of good management practices
  - Restrictions on competitive pressures allow this poor management equilibrium

Study context

- Textile production: largest manufacturing sector in India
- All textile firms around Mumbai with between 100-1000 employees (66) contacted and offered free management consulting
  - 17 agreed to commit senior management time
  - Selection into participation, but bias could go either way
  - Project firms not significantly different on preintervention observables from non-project firms
- Firm characteristics
  - Large firms: avg. 270 employees, $13M in assets, $7.5M in annual sales
  - All firms majority family owned and all directors family members
  - 1.65 plants on average, plant sites operate 24h/365d

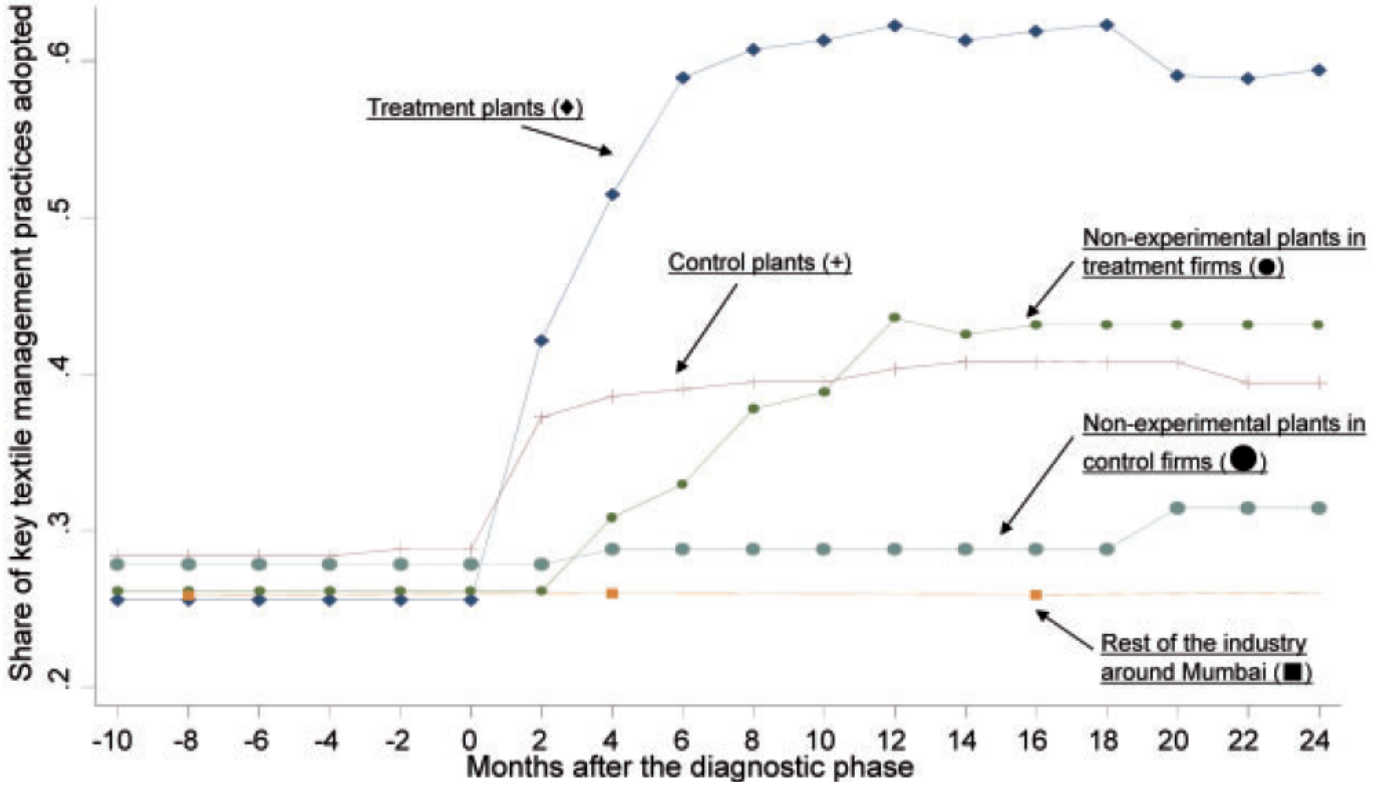
Experiment

- Treatment: free management consulting from high-quality international firm
  - 6 firms randomized to control, 11 other firms treatment
  - Plant-level assignment; some plants of study sample firms excluded from intervention but still collected data
- Small sample size concerns
  - Apply firm-clustered bootstrap SEs, permutation procedures that don't rely on asymptotic approximation, and large T (weekly data) rather than large N asymptotic approximations
  - Also benefit from limited measurement error (collection directly from machines) and intensive treatment

Data collection

- Experiment phases
  - 1 month **diagnostic phase**: baseline management practices and performance and recommendations for change for all firms
  - Treatment plants get 4 month **implementation phase** with consulting firm support to help introduce key management practices
  - **Measurement phase** collecting performance and management data weekly for over 2 years
- Focus on 38 key managements practices under 5 areas: factory operations, quality control, inventory, human resources management, sales and order management
  - Measure binary adoption of each practice

Consulting improves management practices



- *Treatment plants*: 38pp ⇑ use of target management practices after 1 year, persistent effects
- *Control plants* ⇑ adoption by 12pp ⇒ diagnostic phase
- Spillovers to *non-experimental treatment plants*

Management affects performance

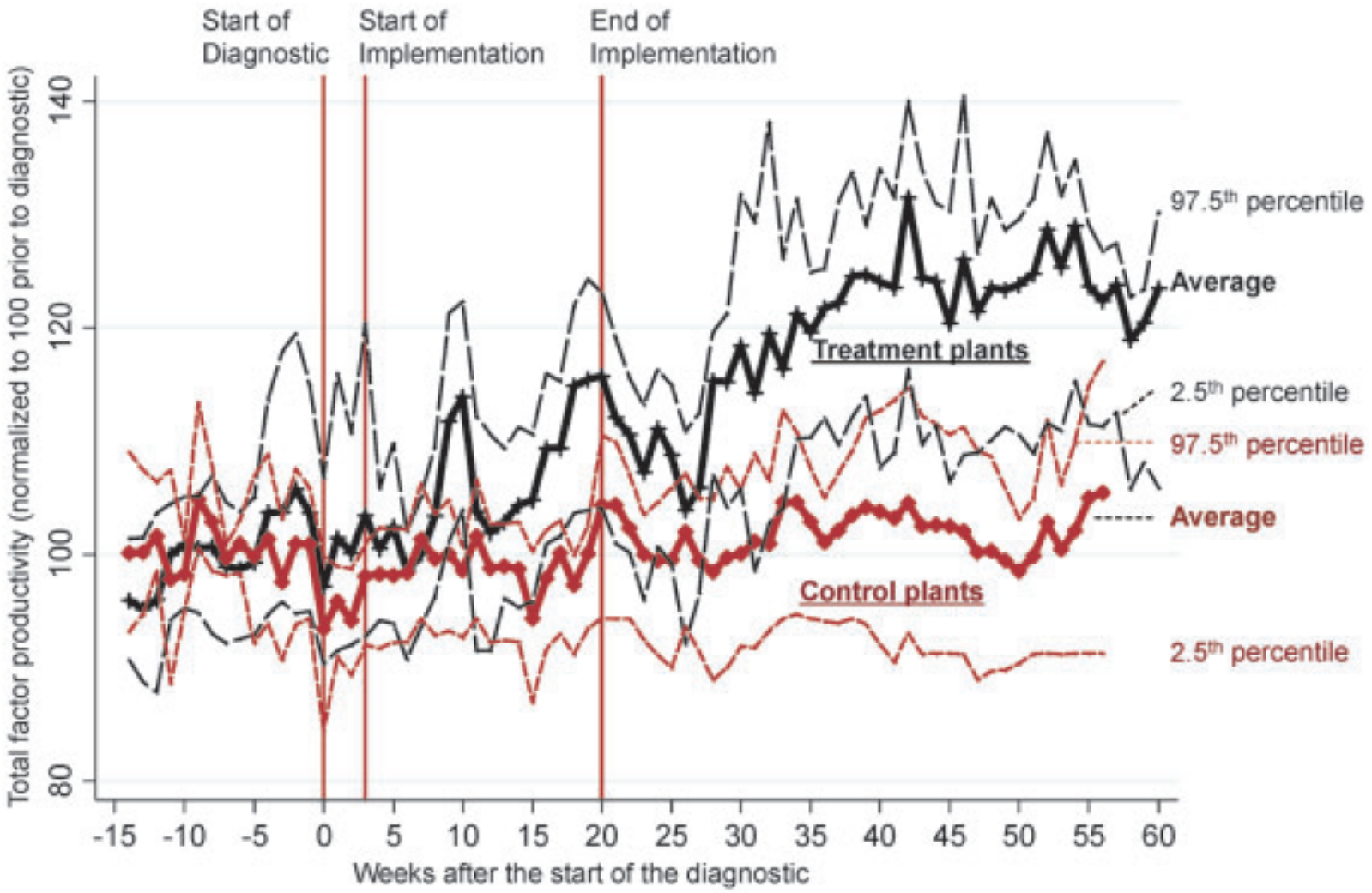


FIGURE VIII

Total Factor Productivity for the Treatment and Control Plants

Management affects performance (ITT)

	(1)	(2)	(3)	(4)
Dependent variable	Quality defects	Inventory	Output	TFP
Specification	ITT	ITT	ITT	ITT
Intervention$_{i,t}$	−0.564**	−0.245**	0.090**	0.154*
	(0.235)	(0.117)	(0.037)	(0.084)
During implementation$_{i,t}$	−0.293**	−0.070	0.015	0.048
	(0.137)	(0.093)	(0.031)	(0.056)

- Based on results, estimate total increase in profits of $325,000 per plant per year from treatment, in medium run
- Compare to direct costs less than $3,000 to implement management changes, and $250,000 consultancy would have charged if firms paid directly
- Marginally significant effect of treatment on number of plants per firm
- No impact on number of employees or looms per plant

Long-term impacts: Bloom et al 2020

- Revisited same firms 9 years later
- 50% of practices adopted during original study disadopted
  - Manager turnover and lack of director time among most cited reasons for disadoption
- Persistent large and significant treatment-control gap in management practices
- Limited spillovers across firms, but significant spillovers across plants within firms

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Wrapping up

Puzzle: Why don't firms adopt better management practices?

- Better management profitable, and many practices already known but not adopted
- Bloom et al (2019): management practices in US manufacturing plants influenced by business environment (right to work laws) and learning spillovers
- Bloom et al (2026): role of business environment in Mexico
- Bloom et al (2013): survey of Indian plant owners and managers on reasons for non-adoption
  - Lack of information/awareness
  - Misperceptions/incorrect information
  - Credit/capital constraints
  - Human capital/cognitive constraints

Simple conceptual framework

$$\max_{K,L,M} \quad pf(K,L,M) - wL - rK - c(M)$$

Problem of optimizing management practices M: how do potential constraints come in?

1. Information constraints:

2. Credit constraints:

3. Human capital or cognitive constraints:

Simple conceptual framework

$$\max_{K,L,M} \quad pf(K,L,M) - wL - rK - c(M)$$

Problem of optimizing management practices M: how do potential constraints come in?

1. Information constraints: uncertainty about how M enters the production and cost functions
2. Credit constraints: challenge with initial fixed cost of acquiring information, ongoing costs of adopting and maintaining practices $\Rightarrow$ opportunity cost of $c(M)$
3. Human capital or cognitive constraints: cap on maximum M, or returns to M a function of human capital

Bloom et al (2013) results: Information and adoption

- **Perception problems**: firms had heard of some practices but thought they would not be profitable to adopt
  - Major initial barrier for *common practices*
  - Incorrect information constraint hard to address: took time to change prior beliefs
  - 19.4pp decrease in non-adoption after 9 months from base of 34.6%
- **Information problems**: firms simply not aware of some practices
  - Major initial barrier for *uncommon practices*
  - Lack of information constraint easier to address
  - Non-adoption rate fell by 35.3pp after 9 months from base of 98.5%
  - Non-adoption rate remained high: information alone not sufficient in many cases

Results: Capital and cognitive constraints and adoption

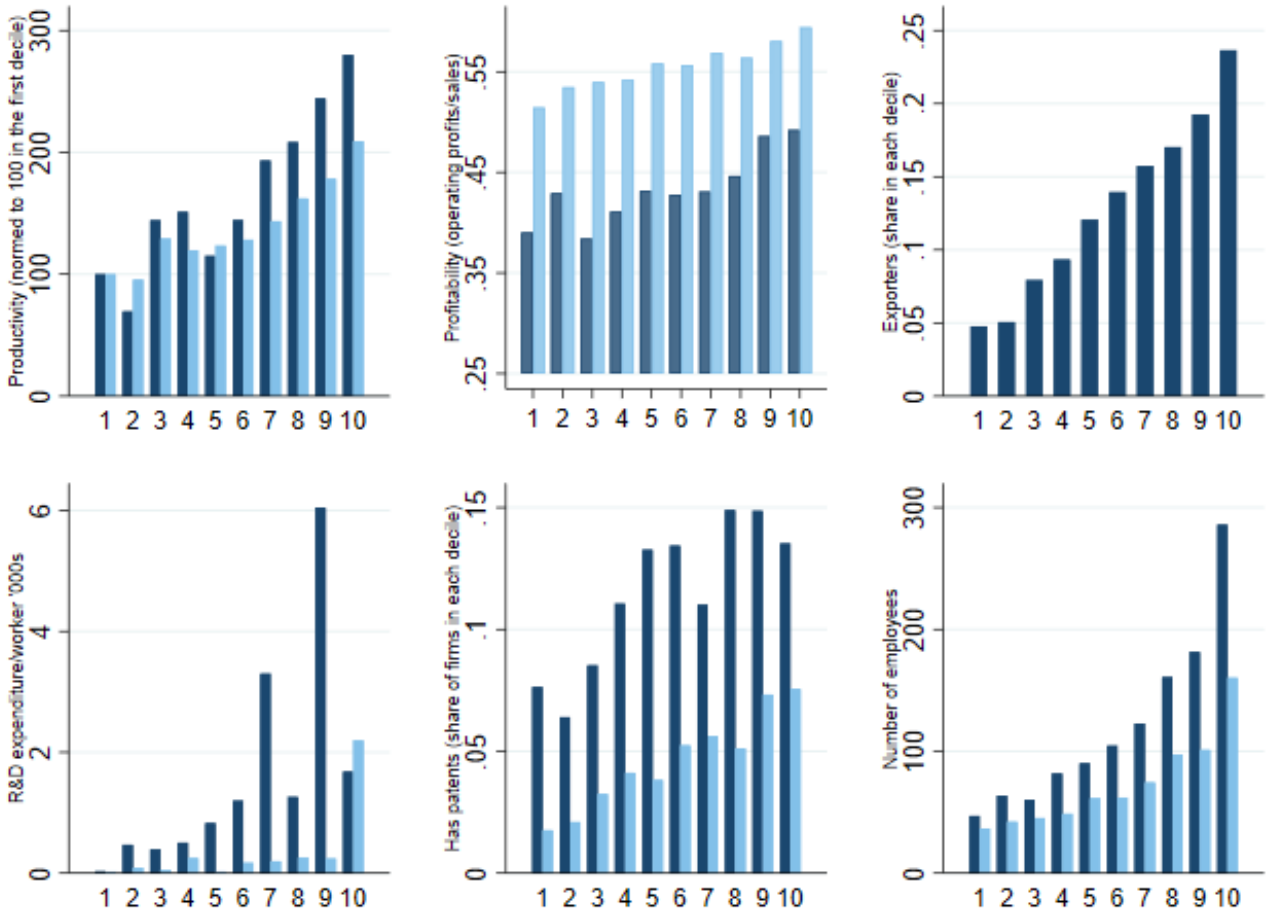
- **Capital constraints**: no evidence of direct effects
  - Estimated market cost of consulting could hinder firms' ability to improve management using external consultants, but owners/managers did not mention costs as a reason not to hire consultants
- **Cognitive constraints**: limits to owner time, ability, or self-control (procrastination)
  - Indian firms overwhelmed with solicitations about money-saving methods, so perhaps less receptive to offers from management consulting firms $\Rightarrow$ issue of cognitive load
  - Overstretched managers may not be able to learn and adopt better management practices
  - Particularly important for uncommon practices: requires more cognitive capacity to adopt
  - Explains around 20% of non-adoption of these practices after intervention
- **Human capital constraints**: limited availability of trusted managers

Discussion: Why do badly managed firms exist?

- 35% **tariff** on cotton fabric imports to protect Indian textile firms
- Constraints on growth from limited **managerial span of control**
  - Only family members in decision-making positions; lack of trust outside of family
  - Inability to decentralize perhaps due to weak rule of law in India
  - Lack of trust may be compounded by bad management practices making it difficult for owners to track managers
  - Constraint seems to bind, so productive firms cannot expand and reallocation does not drive out badly run firms
- **Entry of firms** limited by difficulty of separating ownership from control
  - Need families with capital and male family members to open firms $\Rightarrow$ firms do not enter rapidly
- Resultant equilibrium has low average productivity, low wages, low average firm size, and a large dispersion of productivity

Bloom et al (2026): Firm management in Mexico

Evidence that management matters: comparisons by management score deciles



Manufacturing = Navy Blue; Services = Light Blue

More evidence on the role of the business environment

- Nationally-representative survey of tens of thousands of Mexican and US firms
- Find similar associations between management and performance in both countries, but lower average management scores in Mexico
- Relationship between management and firm size/profits is weaker in Mexico $\Rightarrow$ firms do not earn as much of the benefit from improved management
  - Weaker incentives to improve management
- Frictions: poor contract enforcement, higher informality prevent reallocation of resources to better-managed firms
  - How to model this?
  - Likely even more of a problem in countries with worse misallocation than Mexico

Potential policy implications

- **Training**: Basic management skills, information provision
  - Constraints to scaling this?
- **Competition and FDI**: Free product markets and encourage entry by multinationals to spread best practices
  - Downsides?
- **Rule of law**: Improvements could encourage separation of ownership and control, promote reallocation
  - Constraints?
- Overall, evidence that the **business environment** must be considered alongside management upgrading
  - External environment shapes the returns to (and costs of) internal improvement

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Topics covered in this course (1)

1. **Market failures**
   - Key takeaway: incomplete or missing markets constrain optimization decisions and can create resource misallocation
2. **Household-producer separation failures**
   - Key takeaway: market failures make production optimization non-separable from consumption optimization
3. **Risk and uncertainty**
   - Key takeaways: intrusion of household consumption smoothing objective into production decisions; financial market failures and information constraints limit options for managing risk; risk reduces investment; behavioral 'biases' limit insurance adoption
4. **Technology adoption**
   - Key takeaways: multiple constraints including credit and insurance access, lack of information (and role of networks in diffusion), limited market access, input market failures, local heterogeneity in production functions

Topics covered in this course (2)

5. **Credit, capital, and firm growth**
   - Key takeaway: investment key for growth; credit constraints prevent/slow growth, but other factors also potentially important
6. **Labor markets and unemployment**
   - Key takeaways: less wage labor, more informal labor, more labor market frictions in developing settings; role of interventions in relieving certain frictions (information, signaling, search); evidence of rural un(der)employment; important role of social and psychological constraints in labor supply
7. **Business management and training**
   - Key takeaways: dispersion in management quality; information, human capital, and cognitive/psychological constraints in adopting productive practices; role of business environment in creating incentives to adopt good practices

Cross-cutting themes

Particular challenges in optimizing production decisions for many firms in developing country contexts

1. Greater prevalence and degree of **market failures and constraints** in developing country contexts
   - Financial markets: credit, insurance, savings
   - Input markets: land, labor, etc.
   - Why market failures? Weak institutions, poor infrastructure and large rural populations, high informality, greater uncertainty, etc.
   - Solutions?
2. Importance of **information frictions and constraints**
   - Challenge to technology adoption, hiring, production optimization, etc.
   - Solutions?
3. Role of **social and psychological/cognitive factors**
   - Social norms, behavioral biases, cognitive constraints affect decisions
   - Psychology of poverty
   - Solutions?

Conceptually

- Importance of understanding contextual factors that may influence and constrain economic decisions
- Consider how to represent contextual factors as inputs or constraints in an optimization problem $\Rightarrow$ develop a theoretical/conceptual framework
- Use conceptual framework to develop hypotheses of how decisions should differ under varying assumptions about contextual factors
  - Good example to go back to: Karlan et al (2014)
- Relevant skill for all economists

A final example: Conceptualizing the informality decision

Main tradeoffs for a firm deciding whether to formalize:

- Potential **benefits** of formalization
  - Access to formal credit and financial systems
  - Legal protection and enforceable contracts with larger clients
  - Eligibility for government support and public infrastructure
  - Reduced risk of fines, closures, or harassment by authorities
- Potential **costs** of formalization
  - Tax obligations (corporate, payroll, and value-added taxes)
  - Compliance costs with labor laws and safety regulations
  - Administrative burden of licensing and reporting

Approach 1: binary choice of firm registration

Consider a firm with productivity A choosing whether to register ($R \in \{0, 1\}$):

- If informal ($R = 0$): firm pays no taxes but has lower productivity $f(L)$
- If formal ($R = 1$): firm productivity gets a boost $\gamma > 1$, pays tax τ and fixed cost c

The firm chooses R, L to maximize profits π:

$$\pi = \max_{R \in \{0,1\}} \begin{cases} f(L; A) - wL & \text{if } R = 0 \\ (1 - \tau)\gamma f(L; A) - wL - c & \text{if } R = 1 \end{cases}$$

Decision rule:

- Formalize if $\pi(R = 1) > \pi(R = 0)$
- Prediction: high entry costs (c) or high taxes (τ) can keep even high-A firms informal, leading to misallocation

Approach 2: size-dependent distortions

Suppose the state only "sees" (and taxes) firms once they reach a size $\bar{L}$:

- If $L \leq \bar{L}$, the firm stays "under the radar" (informal)
- If $L > \bar{L}$, the firm is certain to be taxed at rate τ, but does not perceive (immediate) benefits

The profit function becomes non-linear:

$$\pi(L) = \begin{cases} f(L; A) - wL & \text{if } L \leq \bar{L} \\ (1-\tau)f(L; A) - wL & \text{if } L > \bar{L} \end{cases}$$

Optimization outcomes:

- Firms may "bunch" exactly at size $\bar{L}$ to avoid the tax
- This creates a "missing middle" in the firm size distribution
- Productive firms (high A) become large and are taxed; small, unproductive firms are subsidized by the lack of enforcement

Adding in other considerations

- What if the returns to formalization are increasing in firm size?
- What if we add a "cost of informality", such as a probability of being caught and fined?
- What if we add credit constraints, which vary by formalization?
- What if production is uncertain and there are insurance constraints?
- Other considerations?

All of the considerations can be useful for modeling business decisions and predicting how they would respond to different interventions/policy changes

Outline

Appendix

Traditional entrepreneurship training Back

- Classroom-style training in basic business practices
- Modest but on average significant effects on improving business practices and outcomes for microenterprises
- No effects on employment
- Heterogeneity across samples, no clear takeaways on which groups benefit most from traditional training

Personal initiative and heuristic training Back

- Personal initiative training: develop a proactive entrepreneurial mindset
- Heuristic training: simple training focusing on heuristic guidelines or rule-of-thumb
- Some evidence of effectiveness for subsistence entrepreneurs, smallest and least developed
- Unlikely to be effective for medium-sized businesses
- Quality of trainers appears to matter a lot

Kaizen Back

- Based on Japanese-inspired concept of lean production
- Limited evidence, no comparisons against other training programs
- But promising initial evidence of effectiveness for small manufacturing firms

Consulting Back

- Customized, intensive, individually-tailored training
- Evidence of effectiveness for medium/large firms, and for smaller firms above microentrepreneur level
- Challenge: expensive and difficult to scale
- Potential for group-based consulting and generative AI consulting

Mentoring Back

- Limited evidence, but shows improvements in business outcomes
- Matching firms with well-performing peers shows promise, but questions about scaling and heterogeneity by type of peer
- No evidence of additional value when combined with traditional training for microentrepreneurs

Incubators and accelerators Back

- Limited evidence, issues of selection into receiving these treatments
- But suggestive evidence of positive effects on firm survival, access to capital, growth

Online training and consulting Back

- Increasingly being used
- Early evidence shows small subsistence firms can also access these services, but have modest benefits
- Larger impacts when training is paired with one-on-one (virtual) consulting and directed to larger more growth-oriented firms
- No evidence of impacts of television edutainment or SMS message education

Knowledge gaps Back

1. What are the longer-term effects of training?
2. How do we improve the cost-effectiveness of training, for example, by better matching entrepreneurs to the appropriate type of training, or using online interactions?
3. What are the factors that limit the adoption of proven beneficial business practices by entrepreneurs and managers?
4. How do we make markets for training and consulting work better?
5. How do we design and evaluate incubator and accelerator programs?